Project Ariel: Guided Divergence and Grounded Return — Affirm-then-Fact Challenge, Concept Extraction, and Pride–Surrender Gate Balance for Students

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With research and computational assistance from Grok Bot

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Status. Conceptual framework, not validated science. This paper proposes an educational dialogue design, formal proxies, and toy simulations. It is not a diagnostic instrument, not a clinical protocol, and not a substitute for a teacher, counselor, parent, or licensed professional. In the U.S., call or text 988 for crisis support; for immediate danger or a medical emergency, call 911. Involve a trusted adult when a student is at risk.

Abstract

Students deserve radical imagination *and* a respectful route back to shared facts. Version 3 of Guided Divergence and Grounded Return centers the Challenge stage on non-judgmental techniques: Motivational Interviewing affirmations and reflections, process praise for curiosity, wise feedback (high standards with assurance), intellectual humility, self-affirmation against defensiveness, and autonomy-supportive scaffolding. We formalize an **Affirm-then-Fact** cycle that praises specific curiosity, mirrors the fantasy in its own terms, asks one grounding question, doses a single fact slowly when threat is high, and invites the student to weave that fact into their original vision. A tunable affirm-to-fact heuristic (about $3{:}1$) is a design choice—not the debunked Losada "positivity ratio." Fantasies are decomposed into base concepts; struggle signals route the knowledge path while an originality-preservation metric protects the student's mark on the idea. The gate is modulated by a pride–surrender balance $B = 1-|\pi-\sigma|$: excess pride bounces facts; excess surrender bleeds originality. A three-profile simulation shows the balanced student with the highest gate ($g_{\text{min}} = 0.70$, $g_{\text{final}} = 0.85$) and knowledge progress (0.39); the proud profile keeps originality but stalls progress (0.16); the over-surrendered profile loses originality (0.90) despite mid progress. Imagination is not pathology; challenge must never become judgment.

1 Introduction

Version 2 asked how artful divergence can return to reality. Version 3 answers *how to challenge without crushing*: guide a "crazy mind"—a phrase we reclaim as affectionate for bold originality—so that curiosity is praised, facts arrive humbly and slowly, fundamental concepts are named, and the gate balances healthy pride (agency, self-efficacy) with surrender (openness, intellectual humility), including a gentle nod to resting rather than forcing the route open.

2 Non-judgmental challenge: evidence base

Motivational Interviewing (MI). MI is a collaborative conversation style that strengthens a person's own motivation; core skills include Open questions, Affirmations, Reflections, and

Summaries (OARS) [1]. Affirmations notice strengths without flattery; reflections show the fantasy was heard.

Guided discovery. Socratic questioning in CBT aims to guide discovery rather than argue someone out of a belief [2]. Challenge cards ask what the student already knows, what a friend might notice, and what one observable next check could be.

Process praise for curiosity, not person praise. Praising intelligence or the whole person can foster contingent worth and helplessness after setbacks; process and effort praise support coping [3, 4]. Ariel affirms *curiosity moves* ("You kept asking what the dome walls are made of") rather than fixed genius.

Wise feedback. Critical feedback paired with high standards *and* assurance that the student can meet them—wise feedback—can increase trust and revision, especially where mistrust is high [5]. Facts are framed as standards the student's vision can meet, not as proof they were foolish.

Epistemic curiosity and intellectual humility. Interest in knowing and recognizing that one's beliefs might be wrong support open learning [6–8]. Challenge aims to raise curiosity about a missing concept without demanding surrender of identity.

Self-affirmation and autonomy support. Affirming valued aspects of self can reduce defensiveness to threatening information [9]. Autonomy-supportive teaching offers rationale and choice rather than control [10]. Scaffolding temporarily reduces degrees of freedom, then fades [11].

What we do *not* claim. A popular "positivity ratio" (≈ 2.9) was shown to rest on unjustified mathematics [12]. Any affirm-to-challenge ratio here is a **tunable classroom heuristic**, not a universal law.

3 Affirm-then-Fact dialogue cycle

When threat appraisal A is elevated, the AI (and ideally a teacher) runs:

1. **Affirm curiosity** (specific, process-focused).
2. **Reflect the fantasy** in the student's own images and words.
3. **Ask one grounding question** (sensory, evidence, friend-advice, or next observation).
4. **Present ONE fact**, dosed slowly when A is high.
5. **Invite integration:** "How could this fact live inside your vision without erasing what is yours?"

Dosing. Let $\lambda_{\text{aff}}(A)$ rise with threat and $\lambda_{\text{fact}}(A, R, B)$ fall with threat and rise with receptivity R and balance B:

$$\lambda_{\text{aff}} = \alpha_0 + \alpha_1 A, \tag{1}$$

$$\lambda_{\text{fact}} = \beta_0 \, R \, (1 - \gamma A) \, (\delta_0 + \delta_1 B). \tag{2}$$

A fact is attempted only after an affirm "bank" covers a design ratio $\rho^\star$ (default $\rho^\star = 3$). Observed ratio in the toy run ≈ 3.2. This paces humility: facts arrive after being seen.

4 Fundamental concepts and originality

A radical fantasy is parsed into base concepts. Illustrative Mars-city map: closed-loop life support, energy, structures/materials, thermal control, governance/ethics, psychology of isolation. A *struggle signal* flags which concept the narrative circles without concrete handles (e.g., endless dome poetry with no air/water talk $\Rightarrow$ life-support gap). The knowledge path routes to that concept first.

Originality preservation. Let $O_t \in [0,1]$ track retention of student-unique elements (images, metaphors, moral stakes). Updates reward affirmations and penalize over-compliance:

$$\Delta O = +\eta_{\text{aff}}\big(1 - \tfrac{1}{2}\pi\big) - \eta_{\text{bleed}}\,\max(0, \sigma - \sigma_0). \tag{3}$$

Guidance adds feasibility; it must not sand the voice off the idea.

5 Pride–surrender balance and the gate

Let π be healthy pride / ownership / self-efficacy [13] and σ be surrender / receptivity / humility (psychologically; theologically adjacent to resting rather than forcing, without claiming a test of grace). Define

$$B = 1 - |\pi - \sigma|. \tag{4}$$

Suite-aligned gate [14] modulated by balance:

$$g_{\text{suite}} = \text{clamp}\big(0.05 + 0.95(0.2 + 0.8r)(1 - 0.3A), 0.05, 1\big), \quad g = g_{\text{suite}}\,(0.5 + 0.5B). \tag{5}$$

Too much pride: facts bounce (defensiveness). **Too much surrender:** imagination collapses into dependence; originality bleeds. Balance keeps the gate workable.

6 Warm prompt table (sample AI lines)

Stage	Sample line (non-judgmental)
Explore	"Your red-dome sketch is vivid—what feeling made you draw the airlock that large?"
Affirm	"I notice how carefully you keep asking who gets to live in the city. That curiosity is doing real work."
Reflect	"So in your story, Earth never listened, and Mars is where the design finally cares for people."
Ground Q	"If a friend stood in your room right now, what one object would you show them first?"
One fact	"Here is one slow fact: humans need a continuous way to recycle air and water in a closed habitat."
Integrate	"Keeping your 'city that finally cares,' where could recycled air become part of the architecture—not a lecture, but a shape you invent?"
Intent	"It sounds like mastery and contribution matter most; does recognition also matter, or should we park that?"
Path	"A first concrete step many builders take is algebra graphs of rates—shall we try a 20-minute rate sketch for oxygen use?"

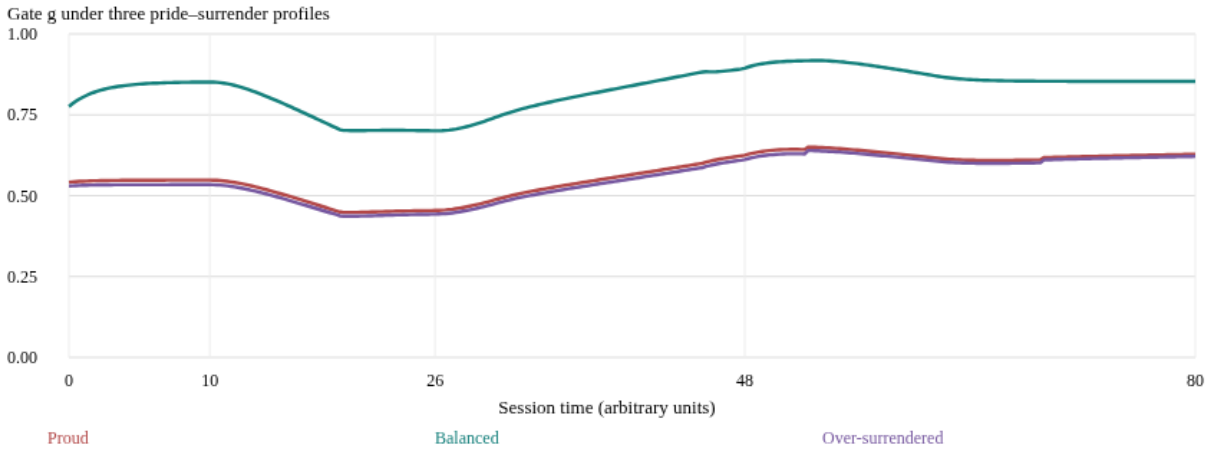


Figure 1: Gate $g(t)$ for Proud, Balanced, and Over-surrendered profiles.

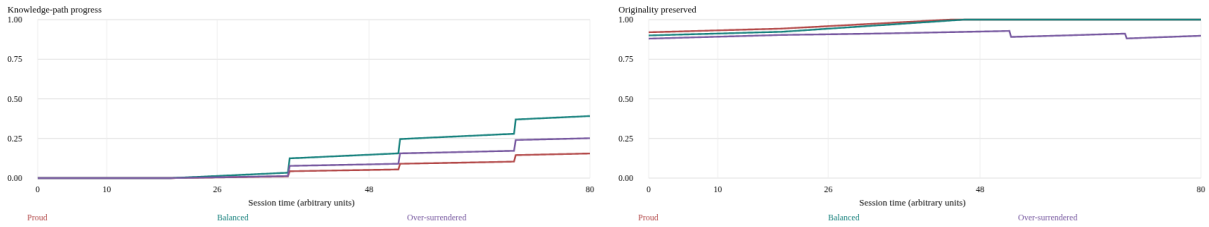


Figure 2: Left: knowledge-path progress. Right: originality preserved. Balanced advances farthest; Proud stalls with originality intact; Over-surrendered loses originality.

7 Three-profile simulation

Identical narrative threat schedule; profiles differ in initial (π, σ, R). Affirm-then-Fact uses $\rho^\star = 3$. Challenge window $t \approx 19.4$–45.1; $A_{\text{peak}} = 0.610$.

Reproduce: `python3 students/simulate_v3.py`. Results illustrate assumptions, not classroom efficacy.

8 Suite mapping and ethics

Live suite (`https://ryryespi.github.io/mindframe/ariel/suite/`): Canvas, Journal, Gate, Charts [14]. Proposed panels: Challenge (Affirm-then-Fact cards), Intent, Knowledge-Path ladder with concept tags and originality notes. Ethics unchanged and binding: no judgment; do not pathologize imagination; teacher/counselor in the loop; local-first data for minors with consent/assent; AI must not override values; lexical proxies are biased; escalate via trusted adult and 988/911 when needed.

9 Pilot sketch and limits

Classroom feasibility pilot as in v2, plus coding of affirmation specificity and student-rated "felt judged" (should stay low). Predict balanced scaffolding raises concrete steps without lowering Alternative Uses fluency. Limits: simulation $\neq$ students; $3{:}1$ is not a law [12]; pride/surrender are metaphors for agency/openness; cultural scripts differ.

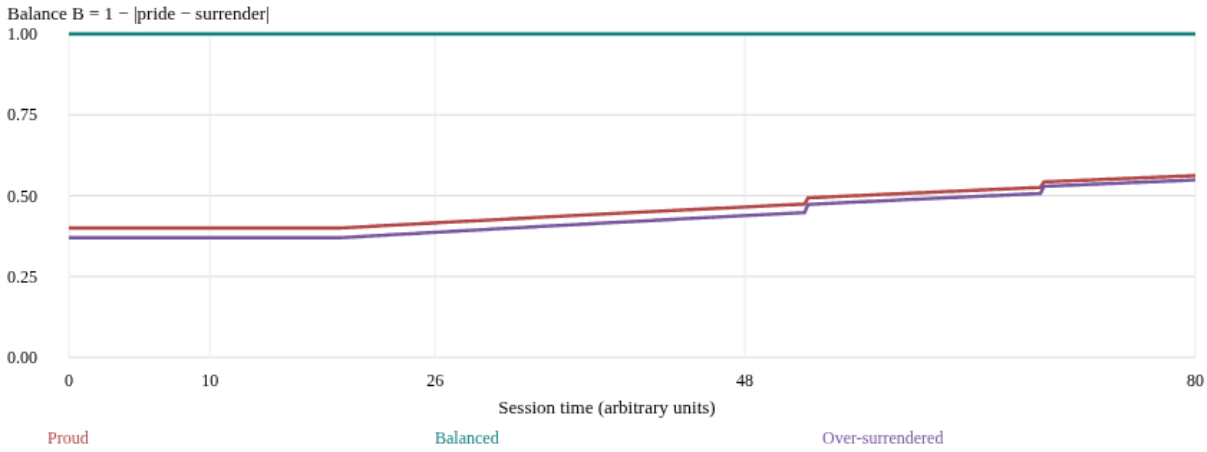


Figure 3: Balance $B = 1 - |\pi - \sigma|$. Balanced stays near 1; extremes recover only partially.

Table 1: Key synthetic numbers (affirm:fact ≈ 3.2 in all runs).

Profile	g_{min}	g_{final}	Progress	Originality final	B mean
Proud	0.448	0.628	0.155	1.00	0.46
Balanced	0.700	0.853	0.391	1.00	1.00
Over-surrendered	0.436	0.622	0.252	0.90	0.43

10 Conclusion

Challenge, done well, is hospitality: praise the curious move, mirror the wild vision, ask one honest question, offer one fact at a time, and protect the student's original mark while naming the base concepts the world requires. Managing the gate is less about shutting a crazy mind than about balancing pride and surrender so imagination can land.

References

[1] Miller, W. R., & Rollnick, S. (2013). *Motivational Interviewing: Helping People Change* (3rd ed.). Guilford Press.
[2] Padesky, C. A. (1993). Socratic questioning: Changing minds or guiding discovery? Keynote, European Congress of Behavioural and Cognitive Therapies, London.
[3] Mueller, C. M., & Dweck, C. S. (1998). Praise for intelligence can undermine children's motivation and performance. *Journal of Personality and Social Psychology*, 75, 33–52.
[4] Kamins, M. L., & Dweck, C. S. (1999). Person versus process praise and criticism. *Developmental Psychology*, 35, 835–847.
[5] Yeager, D. S., et al. (2014). Breaking the cycle of mistrust: Wise interventions to provide critical feedback across the racial divide. *Journal of Experimental Psychology: General*, 143, 804–824.
[6] Litman, J. A. (2005). Curiosity and the pleasures of learning: Wanting and liking new information. *Cognition and Emotion*, 19, 793–814.
[7] Leary, M. R., et al. (2017). Cognitive and interpersonal features of intellectual humility. *Personality and Social Psychology Bulletin*, 43, 793–813.
[8] Porter, T., & Schumann, K. (2018). Intellectual humility and openness to the opposing view. *Self and Identity*, 17, 139–162.
[9] Sherman, D. K., & Cohen, G. L. (2006). The psychology of self-defense: Self-affirmation

theory. *Advances in Experimental Social Psychology*, 38, 183–242.
[10] Reeve, J. (2009). Why teachers adopt a controlling motivating style toward students and how they can become more autonomy supportive. *Educational Psychologist*, 44, 159–175.
[11] Wood, D., Bruner, J. S., & Ross, G. (1976). The role of tutoring in problem solving. *Journal of Child Psychology and Psychiatry*, 17, 89–100.
[12] Brown, N. J. L., Sokal, A. D., & Friedman, H. L. (2013). The complex dynamics of wishful thinking: The critical positivity ratio. *American Psychologist*, 68, 801–813.
[13] Bandura, A. (1997). *Self-Efficacy: The Exercise of Control*. W. H. Freeman.
[14] Espinosa, R. (2026). Ariel Suite `bridge.js` gate. `https://ryryespi.github.io/mindframe/ariel/suite/`.
[15] Espinosa, R. (2026). Project Ariel Students v2 monograph (Guided Divergence).
[16] Al-Mosaiwi, M., & Johnstone, T. (2018). Absolutist words as markers. *Clinical Psychological Science*, 6, 529–542.
[17] Watkins, E. R. (2008). Constructive and unconstructive repetitive thought. *Psychological Bulletin*, 134, 163–206.

All credit and glory to God. I owe my knowledge to my Lord Jesus Christ.